

A CAPIF-Aware AI-Native Telco Orchestration Platform for Intent-to-API Planning Across CAMARA and NEF Services

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Abstract—This paper presents the design, implementation, and empirical evaluation of a mini telco orchestration platform that combines 3GPP Common API Framework (CAPIF) integration, CAMARA-aligned service semantics, NEF-backed telecom realization, and large language model (LLM)-assisted intent planning. The system is implemented as a FastAPI-based control plane with a product-style operator interface, embedded OpenAPI console, live CAPIF invoker onboarding and service discovery, provider publication, persistent NEF data services, and guarded orchestration execution. The platform supports both deterministic and LLM-assisted planning modes and validates generated API payloads through a three-layer deterministic validator before execution. Empirical evaluation on a 60-entry bilingual (English/Turkish) ground-truth dataset yields: IWSR=100% (deterministic) and IWSR=95% (LLM-assisted, Claude Haiku 4.5), with SCR=SVR=100% in both modes and a Stability Score of 0.938 over repeated LLM runs. The resulting system provides a practical and measurable baseline for reliability-aware intent-to-API orchestration research in 5G and beyond-5G service exposure environments.

Index Terms—CAPIF, CAMARA, NEF, 5G service exposure, intent-based orchestration, LLM planning, telecom APIs, OpenCAPIF, reliability evaluation

I. INTRODUCTION

Programmable network exposure is becoming a central design principle in modern telecom systems. CAPIF provides a standardised governance and discovery plane for exposed APIs [1], CAMARA defines developer-facing service semantics [2], and NEF-based providers bridge those interfaces to telecom data and network capabilities [4]. At the same time, large language models (LLMs) introduce a new opportunity: translating high-level natural language intents into structured API workflows without requiring the operator to manually compose payloads and endpoint sequences [6], [8].

This project implements a working telco orchestration platform at that intersection. Rather than remaining at the conceptual architecture level, the system integrates live OpenCAPIF onboarding and discovery, live provider publication, live QoD and Location Retrieval execution, an embedded operator interface, and an LLM-assisted planning layer protected by deterministic validation. The platform is evaluated using

six formally defined reliability metrics on a bilingual intent dataset, providing empirical evidence for the feasibility and limitations of LLM-driven API orchestration under telecom-specific constraints.

II. RELATED WORK

Intent-based networking has been explored at the NFV orchestration level by Rafiq et al. [10], who demonstrated deterministic intent-to-descriptor translation across multiple orchestrators, and surveyed by Saraiva de Sousa et al. [12] as a key future direction for adaptive network service management. Coronado et al. [11] identified intent abstraction and AI-native closed-loop control as foundational pillars of zero-touch 6G management.

On the exposure side, Molina Sanchez et al. [3] implemented the CAPIF Core Function as an open-source microservice; Fragkos et al. [4] introduced NEFSim as an open NEF experimentation platform; and Turkovic et al. [5] demonstrated 3GPP-compliant CAMARA QoD and Location API integration in a real CAM testbed, measuring call-flow latency of 230–340 ms.

LLM reliability for structured API output has been benchmarked by Liu et al. [9], who found that even GPT-4 achieves an average structured-output compliance score of only 0.40. Maatouk et al. [6] identified hallucination and output inconsistency as primary risks for LLM deployment in telecom operations. Lee et al. [7] introduced TelAgentBench, a multi-faceted benchmark for LLM-based telecom agents, confirming that multi-turn planning tasks significantly degrade pass rates. Zeng et al. [8] demonstrated near-perfect workflow generation accuracy when LLMs are grounded in structured API descriptions.

To the best of our knowledge, no prior work provides empirical reliability evaluation of LLM-based intent-to-API translation within a live CAPIF-compliant 5G service exposure environment under telecom-specific schema, semantic, and registry constraints.

III. SYSTEM OBJECTIVES AND CONTRIBUTIONS

The platform was built with four primary objectives:

- to unify CAPIF onboarding, discovery, and publication inside one operator-facing control plane,
- to expose CAMARA-aligned telecom capabilities through a live orchestration backend,
- to support both deterministic and LLM-assisted planning for natural-language requests,
- to ensure that execution remains guarded by deterministic validation and schema enforcement.

Its main contributions are:

- a working CAPIF-aware telco orchestration backend,
- a pluggable multi-provider LLM planning layer grounded by local service metadata and CAPIF discovery,
- a three-layer deterministic validator (schema, semantic, registry),
- a live provider integration across QoD, Location Retrieval, QoS Provisioning, and QoS Profiles,
- six formally defined reliability metrics with a 60-entry bilingual ground-truth evaluation dataset,
- a structured feedback correction loop for bounded iterative repair of invalid LLM outputs.

IV. PLATFORM ARCHITECTURE

A. High-Level Components

TABLE I
IMPLEMENTED PLATFORM COMPONENTS

Component	Responsibility
Operator Interface	Product-style landing page with embedded OpenAPI console
FastAPI Backend	Route handling, orchestration control, validation, execution
Intent Engine	Deterministic keyword-based intent-to-step routing
LLM Planner	Two-phase structured planning under catalog and CAPIF grounding
Multi-Layer Validator	L1 schema, L2 semantic policy, L3 CAPIF registry checks
Feedback Engine	Bounded correction loop (max 3 iterations) for invalid payloads
CAPIF Client	Invoker onboarding, service discovery, provider publication
Provider Layer	QoD, Location Retrieval, QoS Provisioning, QoS Profiles
Persistence Layer	MongoDB-backed NEF data with automatic initialization

B. Repository Structure

TABLE II
KEY REPOSITORY PATHS

Path	Role
backend/main.py	FastAPI entry point and route registration
backend/intent_engine.py	Deterministic planner and field extractor
backend/llm_planner.py	Two-phase LLM planning with RAG grounding
backend/validator.py	Three-layer validation (L1/L2/L3)
backend/feedback_engine.py	Bounded correction loop
backend/capif/client.py	CAPIF onboarding, discovery, publish
backend/evaluation_engine.py	IWSR/SCR/SVR/HR/SS/RRF computation
frontend/index.html	Operator landing page with API console
providers/NEF/	NEF Monitoring Event stack + Mongo persistence
providers/camara-QoD/	CAMARA Quality-on-Demand service
providers/CamaraLocationRetrieval/	CAMARA Location Retrieval service
providers/NEF-QoS/	3GPP AsSessionWithQoS adapter (port 8585)

V. BACKEND ORCHESTRATION DESIGN

A. Core API Surface

TABLE III
MAIN BACKEND ENDPOINTS

Method	Path	Purpose
GET	/health	Runtime health summary
GET	/catalog/services	CAMARA-aligned capability catalog
POST	/orchestrate	Intent-to-execution planning
GET	/capif/status	CAPIF connectivity check
GET	/capif/invoker	Stored invoker identity
POST	/capif/invokers/onboard	Live invoker onboarding
GET	/capif/discover	Live CAPIF service discovery
POST	/capif/publish	Single service publication
POST	/capif/publish/catalog	Publish all catalog services

B. Supported Telecom Capabilities

The platform exposes four CAMARA-standardised orchestration targets: Quality on Demand (QoD) for temporary QoS session creation; QoS Profiles for profile discovery; QoS Provisioning for long-lived QoS assignment via 3GPP AsSessionWithQoS (TS 29.122); and Location Retrieval for device location access through NEF-backed monitoring.

C. Planning Modes

POST /orchestrate supports three modes: *deterministic* (rule-based, always available); *llm-assisted* (requires configured LLM API key, fails explicitly if planning fails); and

auto (attempts LLM first, falls back to deterministic). This keeps the AI layer optional yet operational for both live demonstrations and controlled experimentation.

VI. CAPIF INTEGRATION WORKFLOW

A. CAPIF Roles

CAPIF provides the exposure governance plane. The platform acts as an **Invoker** consuming exposed APIs; CAMARA providers act as **AEFs** publishing service descriptions; and the OpenCAPIF stack provides the **Core Function** managing discovery, authentication, and onboarding [3].

B. Invoker Onboarding Flow

The onboarding implementation in `backend/capif/client.py`: (1) authenticates against the register service; (2) generates an RSA private key and certificate signing request (CSR) locally; (3) submits the onboarding request to CAPIF Core; (4) receives an `apiInvokerId` and invoker certificate chain; (5) stores runtime credentials for subsequent discovery and authenticated execution. This enables the platform to operate as a real CAPIF invoker rather than a static mock client. CAPIF thus serves three distinct functions: **identity onboarding**, **service governance** (APIs are visible only after publication), and **service discovery** for orchestration-time reasoning.

C. Discovery and Publication

`GET /capif/discover` queries the CAPIF registry using the invoker's mTLS certificate and returns live service descriptions. Provider publication follows the standard APF sequence: onboard provider identity, submit OpenAPI metadata, verify appearance in discovery results.

VII. CAMARA AND NEF SERVICE REALIZATION

CAMARA defines the API-facing service contracts (the *semantic layer*), while CAPIF remains the governance and exposure layer. Provider services from the FRONT-research-group [5] bridge these APIs to telecom behavior: the QoD provider handles temporary session creation; the NEF Monitoring Event provider resolves phone numbers to IMSI identifiers, reads location data from MongoDB, enriches responses with polygon area information, and returns a 3GPP-style monitoring result that the `CamaraLocationRetrieval` adapter normalises into the CAMARA Location Retrieval format [4].

The Location Retrieval end-to-end path executes as: (1) intent mapped to `location-retrieval`; (2) provider constructs a 3GPP `MonitoringEvent` subscription; (3) provider obtains CAPIF security token; (4) NEF API is called; (5) NEF resolves identifiers and polygon data; (6) response normalised to CAMARA format and returned.

VIII. LLM-ASSISTED PLANNING LAYER

A. Two-Phase Planning Strategy

`backend/llm_planner.py` implements a two-phase pipeline. *Phase 1* (service selection): the LLM receives

the user intent, a RAG-injected summary of all CAMARA services and operations, and optionally live CAPIF discovery results; it returns a JSON object specifying `service_id`, `operation_id`, `method`, `path`, and `rationale`. *Phase 2* (payload generation): given the selected operation, the LLM receives the full CAMARA OpenAPI schema for that operation (parsed from YAML at runtime) and returns a complete, schema-aligned JSON payload. The planner proposes structured actions but never executes them directly; all output passes through the three-layer validator before dispatch.

Three LLM providers are supported via a unified dispatcher: Gemini (`GEMINI_API_KEY`), OpenAI (`OPENAI_API_KEY`), and Anthropic (`ANTHROPIC_API_KEY`). The full 60-entry evaluation in this paper used **Claude Haiku 4.5** (Anthropic, model: `claude-haiku-4-5-20251001`, temperature 0.1) at a total API cost of approximately \$0.35.

B. Structured Feedback Correction Loop

When Phase 2 output fails validation, the feedback engine constructs a correction prompt containing the original intent, the rejected payload, and layer-tagged error messages (e.g., [L1] `device.phoneNumber` must match E.164), and resends it to the same LLM. This loop repeats up to `MAX_ITERATIONS = 3`. The recovery result is recorded in `FeedbackMetadata` and returned in the orchestration response.

C. Representative Orchestration Result

An LLM-assisted dry-run request with intent “*Boost video QoS for +905551112233 for 15 minutes*” produced: `service=qod`, `operation=createSession`, `path=/quality-on-demand/v1/sessions`, `duration=900s`, `profile=QOS_E_STREAMING`. The three-layer validator confirmed structural, semantic, and registry compliance in a single pass.

IX. RELIABILITY EVALUATION FRAMEWORK

A. Formal Metric Definitions

Intent-to-API translation is formalised as a mapping $f_\theta : \mathcal{I} \rightarrow \mathcal{W}$, where $\mathcal{I}$ is the set of natural language intents and $\mathcal{W}$ is the set of CAPIF-compliant API workflows generated by LLM f_θ .

1) Intent-to-Workflow Success Rate (IWSR):

$$\text{IWSR} = \frac{|\{w \in \mathcal{W} \mid w = w^*\}|}{|\mathcal{I}|} \quad (1)$$

where w^* is the ground-truth workflow. IWSR measures end-to-end correctness: correct service selection AND valid schema AND valid telecom semantics.

2) Schema Compliance Rate (SCR):

$$\text{SCR} = \frac{|\{w \in \mathcal{W} \mid w \models \text{Schema}\}|}{|\mathcal{W}|} \quad (2)$$

Fraction of generated payloads passing Layer 1 (Pydantic) validation.

3) Semantic Validity Rate (SVR):

$$\text{SVR} = \frac{|\{w \in \mathcal{W} \mid w \models \text{TelecomPolicy}\}|}{|\{w \in \mathcal{W} \mid w \models \text{Schema}\}|} \quad (3)$$

Among schema-compliant payloads, fraction also passing Layer2 telecom policy checks (E.164 format, valid QoS profile, duration bounds, etc.).

4) Hallucination Rate (HR):

$$\text{HR} = \frac{|\{w \in \mathcal{W} \mid w \not\subseteq \mathcal{A}_{\text{reg}}\}|}{|\mathcal{W}|} \quad (4)$$

where $\mathcal{A}_{\text{reg}}$ is the set of APIs discovered via CAPIF. Measures service-selection hallucinations (service does not exist in the CAPIF registry).

5) *Stability Score (SS)*: Given k repeated runs producing d distinct output workflows:

$$\text{SS} = 1 - \frac{d-1}{k} \quad (5)$$

Perfect stability ($d = 1$, identical output every run) yields $\text{SS} = 1$. Computed by running the same 60-entry dataset 5 times and measuring output variance per entry.

6) Recovery Rate after Feedback (RRF):

$$\text{RRF} = \frac{|\{w_c \mid w_i \not\models \text{Schema} \wedge w_c = w^*\}|}{|\{w_i \mid w_i \not\models \text{Schema}\}|} \quad (6)$$

Among initially-invalid payloads, fraction successfully corrected by the feedback loop to a valid, ground-truth-matching workflow.

B. Evaluation Dataset

The ground-truth dataset (datasets/ground_truth.json) contains 60 manually crafted entries: 20 for QoD, 15 for Location Retrieval, 10 for QoS Profiles, 10 for QoS Provisioning, and 5 edge cases; difficulty split: 24 easy, 20 medium, 16 hard; language split: 32 English, 28 Turkish (50/50 bilingual coverage). Phone numbers span four country codes (+30 GR, +90 TR, +34 ES, +1 US) from FRONT-research-group test fixtures.

C. Evaluation Results

TABLE IV
RELIABILITY METRICS: DETERMINISTIC VS. LLM-ASSISTED (60 ENTRIES)

Metric	Deterministic	LLM (Claude Haiku 4.5)
IWSR	100.0%	95.0%
SCR	100.0%	100.0%
SVR	100.0%	100.0%
HR	0.0%	5.0%
SS	1.000	0.938
RRF	N/A	N/A ^a
Avg. latency	<1 ms	≈3 s

^aRRF is undefined when all payloads pass schema validation on first attempt (SCR = 100%; feedback loop was never triggered).

TABLE V
IWSR BREAKDOWN (LLM-ASSISTED MODE)

Dim.	Category	N	IWSR
Language	English	32	93.8%
	Turkish	28	96.4%
Difficulty	Easy	24	100.0%
	Medium	20	100.0%
	Hard	16	81.2%
Service	QoD Session	20	90.0%
	Location Retrieval	15	100.0%
	QoS Profiles	10	100.0%
	QoS Provisioning	10	100.0%
	Edge Cases	5	80.0%

Key findings: (1) SCR and SVR are perfect in both modes — when the LLM selects the correct service, it always generates a structurally and semantically valid payload. (2) All 5% IWSR failures in LLM mode are service-selection errors (HR=5%), not payload errors, confirming the effectiveness of Phase 2 schema injection. (3) Hard-difficulty failures stem from deliberately ambiguous intents (e.g., “Apply critical communications QoS”) where the LLM correctly identifies the QoS priority but misroutes to permanent provisioning instead of a temporary QoD session. (4) Turkish and English intents yield no statistically significant IWSR difference, validating bilingual robustness.

X. PERSISTENCE AND RELIABILITY ENGINEERING

A. Persistent NEF Data

The NEF stack stores location and polygon records in MongoDB. To prevent data loss across container restarts, the deployment uses a named Docker volume at /data/db and an auto-seeding initialisation script mounted at /docker-entrypoint-initdb.d/mongo-init.js. The first deployment seeds required data automatically; subsequent restarts preserve the existing dataset without manual reseed.

B. Three-Layer Validation

Execution never depends solely on LLM output. All generated payloads pass three independent validation tiers before any provider call is made:

- **L1 — Schema (SCR)**: Pydantic model validation enforcing required fields, types, and structural nesting.
- **L2 — Semantic Policy (SVR)**: Eight domain-specific rules including E.164 phone format, valid QoS profile identifiers, duration bounds (1–7200 s), IPv4 unicast validation, and operation/method consistency.
- **L3 — CAPIF Registry (HR)**: Live CAPIF discovery query to verify the selected service exists in the registry (non-blocking if CAPIF is unreachable).

This separation between probabilistic LLM reasoning and deterministic telecom enforcement is the central architectural property enabling measurable reliability guarantees.

XI. CONCLUSION

This paper presented a working AI-native telco orchestration platform that integrates 3GPP CAPIF governance, CAMARA service semantics, NEF-backed telecom realization, three-layer deterministic validation, a structured feedback correction loop, and multi-provider LLM-assisted planning into one deployable control plane. Empirical evaluation on a 60-entry bilingual ground-truth dataset demonstrates IWSR=95% for LLM-assisted mode (Claude Haiku 4.5) and IWSR=100% for deterministic mode, with perfect schema and semantic compliance in both cases. All LLM failures are service-routing errors on deliberately ambiguous hard-difficulty intents; no payload schema violations occur. The six reliability metrics (IWSR, SCR, SVR, HR, SS, RRF) constitute a reusable evaluation framework for any LLM-driven API orchestration system. The platform source code, ground-truth dataset, and evaluation scripts are publicly available at <https://github.com/doadurak/mini-telco-platform>.

Future work includes: evaluation with additional LLM providers and model sizes to establish cross-model reliability bounds; integration of 3GPP ZSM closed-loop control for autonomous network management; user studies on intent clarity and its impact on translation reliability; and extension of the dataset with real operator intent logs for ecological validity.

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